

# Classification of Algebraic Varieties

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# Outline of the talk

## 1 Introduction

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- 2 Irreducible subsets

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# Polynomial equations

Algebraic Geometry is concerned with the study of solutions of polynomial equations; say  $n$ -equations in  $m$ -variables

$$\begin{cases} P_1(x_1, \dots, x_m) = 0 \\ \dots \\ P_n(x_1, \dots, x_m) = 0. \end{cases}$$



# 1 equation in 1 variable

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$$P(x) = a_d x^d + a_{d-1} x^{d-1} + \dots + a_1 x + a_0 \quad a_d \neq 0$$

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- If  $a_i \in \mathbb{C}$ , then by the Fundamental Theorem of Algebra, we may find  $c_i \in \mathbb{C}$  such that  $P(x)$  factors as

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- Therefore a polynomial of degree  $d$  always has exactly  $d$  solutions or roots (when counted with multiplicity).
- If one is interested in solutions that belong to  $\mathbb{R}$  (or  $\mathbb{Q}$ , or  $\mathbb{Z}$  etc.), then the problem is much more complicated, but at least we know that there are at most  $d$  solutions.

# Complex solutions

Throughout this talk I will always look for complex solutions  $(z_1, \dots, z_m) \in \mathbb{C}^m$  to polynomial equations

$$P_1(x_1, \dots, x_m) = 0 \quad \dots \quad P_n(x_1, \dots, x_m) = 0$$

where  $P_i \in \mathbb{C}[x_1, \dots, x_m]$ .

# Projective space

- It is more convenient to work on a compactification  $\mathbb{C}^N \subset \mathbb{P}_{\mathbb{C}}^N$  where  $\mathbb{P}_{\mathbb{C}}^N = \{\mathbb{C}^{N+1} - (0, \dots, 0)\} / \{\mathbb{C}^*\}$  is projective  $N$ -space.

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- We have that  $\mathbb{P}_{\mathbb{C}}^0$  is a point,  $\mathbb{P}_{\mathbb{C}}^1 \cong S^2$  is the Riemann sphere and  $\mathbb{P}_{\mathbb{C}}^N = \mathbb{C}^N \cup \mathbb{P}_{\mathbb{C}}^{N-1}$ .

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- We then replace  $P(x_1, \dots, x_N)$  by the corresponding homogeneous polynomial in  $\mathbb{C}[x_0, \dots, x_N]$ . Eg.  
 $x_1^2 + x_2^3 \Rightarrow x_0 x_1^2 + x_2^3$ .



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 $x_1^2 + x_2^3 \Rightarrow x_0 x_1^2 + x_2^3$ .
- We then have:

# Bezout's Theorem

## Theorem (Bezout's Theorem)

*Let  $P_1(x_0, \dots, x_n), \dots, P_n(x_0, \dots, x_n)$  be  $n$  homogeneous polynomials of degrees  $d_1, \dots, d_n$ . If the set*

$$\bigcap_{i=1}^n \{P_i(x_0, \dots, x_n) = 0\} \subset \mathbb{P}_{\mathbb{C}}^n$$

*is finite, then it consist of exactly  $d_1 \cdots d_n$  points (counted with multiplicity).*

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# Irreducible subsets

- In general a **Zariski closed subset** of  $\mathbb{P}_{\mathbb{C}}^m$  is a set of the form

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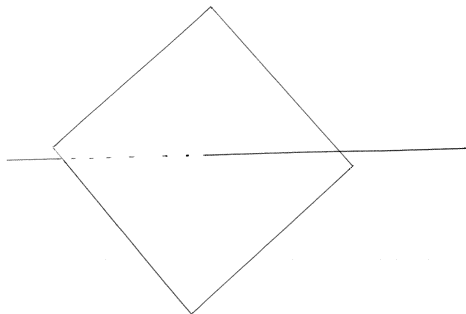
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- If  $P(x, y, z) \in \mathbb{C}[x, y, z]$  is homogeneous of degree  $r > 0$ , then

$$X = \{P(x, y, z) = 0\}$$

is irreducible if and only if  $P(x, y, z)$  does not factor.

## Reducible set

REDUCIBLE ZARISKI CLOSED SUBSET OF  $\mathbb{P}^3$





## Irreducible subsets II

- If  $X$  is an infinite irreducible Zariski closed subset, then there is an integer  $d > 0$  such that for most points  $x \in X$ , one can find an open neighborhood  $x \in U_x \subset X$  where  $U_x$  is analytically isomorphic to a ball in  $\mathbb{C}^d$ .

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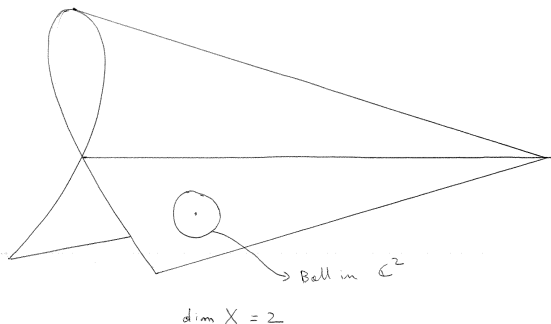
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- We then say that  $x$  is a **smooth** point and that  $X$  has **dimension**  $d$ .
- By a deep result of Hironaka, there exists a **desingularization** of  $X$ , that is a morphism  $f : Y \rightarrow X$ , such that  $Y$  is everywhere smooth and  $f$  is the identity over any smooth point of  $X$ .

# Irreducible set of dimension 1

SINGULAR SET IN  $\mathbb{P}^3_{\mathbb{C}}$



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- The simplest curves are **rational curves** i.e. curves of genus 0.
- There is only one such curve  $X = \mathbb{P}_{\mathbb{C}}^1$ .
- When  $g = 1$  we have a 1-parameter family of **elliptic curves**  $X_t = \{y^2 - x(x-1)(x-t) = 0\}$ .

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- We would like to find a natural (canonical) description for  $X \subset \mathbb{P}_{\mathbb{C}}^m$ .

# Line bundles

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- If  $L$  is a line bundle on  $X$  and  $s_0, \dots, s_m$  is a basis of  $H^0(X, L)$ , then there is a map  $\phi_L : X \dashrightarrow \mathbb{P}_{\mathbb{C}}^m$  given by

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- To define an embedding of  $X$  in  $\mathbb{P}_{\mathbb{C}}^m$ , we must find a line bundle  $L$  whose sections distinguish different points and tangent directions.

# The canonical line bundle

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## Theorem

*Let  $X$  be a curve of genus  $g \geq 2$ , then the sections of  $H^0(X, \omega_X^{\otimes 3})$  define an embedding of degree  $6g - 6$*

$$X \hookrightarrow \mathbb{P}_{\mathbb{C}}^{5g-6}.$$

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- In coordinates this may be described as follows. If  $r_0, \dots, r_m$  are generators of  $R(\omega_X)$  and  $K = \ker(\mathbb{C}[x_0, \dots, x_m] \rightarrow R(\omega_X))$ , then  $X \subset \mathbb{P}_{\mathbb{C}}^m$  is defined by the equations in  $K$ .

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# The canonical ring

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- We have  $\kappa(X) = \max_{k \geq 0} \{\dim \phi_{\omega_X^{\otimes k}}(X)\}$  so that  $\kappa(X) \in \{-1, 0, 1, \dots, d\}$ .

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- The morphism  $\nu$  may be viewed as a surgery that replaces the point  $x$  by a rational curve  $E \cong \mathbb{P}^1 = \mathbb{P}(T_x X)$ .
- The exceptional curve  $E$  is called a  $-1$  **curve** since

$$\omega_{\tilde{X}} \cdot E = \deg(\omega_{\tilde{X}}|_E) = -1.$$

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- It turns out that two surfaces  $X$  and  $X'$  are birational if and only if there is a sequence of blow ups  $X_n \rightarrow X_{n-1} \rightarrow \dots X_1 \rightarrow X$  and  $X'_m \rightarrow X'_{m-1} \rightarrow \dots X'_1 \rightarrow X'$  such that  $X_n = X'_m$ .



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- It is then natural to study surfaces up to birational equivalence.

## Birational geometry of surfaces III

- By a Theorem of Castelnuovo, one may reverse the blowing up procedure i.e. given any surface  $X$  and any  $-1$  curve  $E \subset X$ , there exists a morphism  $\nu : X \rightarrow X_1$  such that  $X = \text{Bl}_x(X_1)$  for some point  $x \in X_1$ .

## Birational geometry of surfaces III

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- By Castelnuovo's Theorem, given any surface  $X$  there exists a birational morphism to a minimal surface  $X \rightarrow X_{\min}$  (given by finitely many blow downs).

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- If  $\kappa(X) \geq 0$ , then the minimal model  $X \rightarrow X_{min}$  is unique and  $\omega_{X_{min}}$  is **nef**. This means that for any curve  $C \subset X_{min}$  we have

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- It then follows that  $R(\omega_X)$  is finitely generated.
- Conclusion: After performing a finite sequence of geometrically meaningful operations (contracting  $-1$  curves) we obtain a birational model  $X_{min}$  on which  $\omega_{X_{min}}$  is positive (nef and semiample).

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- $X_{can}$  has Du Val singularities and degree  $25K_{X_{min}}^2$  in  $\mathbb{P}^N$ .
- Therefore, if  $K_{X_{min}}^2 \leq M$  then there are only finitely many families of such surfaces (modulo birational equivalence).

# Outline of the talk

- 1 Introduction
- 2 Irreducible subsets
- 3 Curves
- 4 Surfaces
- 5 **Higher dimensions.**
  - Running the minimal model program
  - Existence and termination
  - Idea of the proof

# 3-folds

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- They showed that if  $X$  is not covered by  $\mathbb{P}^1$ 's, then there exists a finite set of well understood birational maps (known as flips and divisorial contractions)  $X \dashrightarrow X_1 \dashrightarrow \dots \dashrightarrow X_N$  such that  $\omega_{X_N}$  is semiample (i.e.  $\phi_{\omega_{X_N}^{\otimes m}} : X_N \rightarrow \mathbb{P}^N$  some  $m > 0$ ).

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## Theorem (Birkar-Cascini-Hacon-M<sup>c</sup>Kernan)

*Let  $X$  be a variety of general type (i.e.  $\kappa(X) = \dim(X)$ ), Then there exists a finite sequence of flips and divisorial contractions*

$$X \dashrightarrow X_1 \dashrightarrow \dots \dashrightarrow X_N$$

*such that  $\omega_{X_N}$  is semiample.*



## Higher dimensions II

We can also show that:

Theorem (Birkar-Cascini-Hacon-M<sup>c</sup>Kernan, Siu)

*The canonical ring*

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It follows that there is a natural map  $\phi : X \dashrightarrow \text{Proj} R(\omega_X) \subset \mathbb{P}_{\mathbb{C}}^m$ . In most cases,  $\phi$  carries a lot of information about  $X$  and it can be used to study the geometry of  $X$  in a coordinate free manner.

## Higher dimensions III

One can also generalize some of the features of the classification of surfaces. For example:

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**Theorem (Tsuji, Hacon-M<sup>c</sup>Kernan, Takayama)**

*For any integers  $d, M > 0$ , there are only finitely many families of varieties of general type, dimension  $d$  and volume  $\text{Vol}(\omega_X) = (\omega_{X_{\min}})^d \leq M$  modulo birational isomorphisms.*

# Running a MMP

- A **Minimal Model program** is a sequence of geometric operations  $X_i \dashrightarrow X_{i+1}$  that starting from a projective variety  $X$  produces a minimal model (if  $\kappa(X) \geq 0$ ) or a covering family of rational curves (if  $\kappa(X) < 0$ ).

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- If  $\omega_X$  is not nef, then consider the **cone of effective curves**

$$\overline{NE}(X) = \{ \sum r_i C_i \mid r_i \in \mathbb{R}^{\geq 0} \} / \equiv$$

(here  $C \equiv C'$  or  $[C] = [C']$  if for any line bundle  $L$  on  $X$ , we have  $L \cdot (C - C') = 0$ ).

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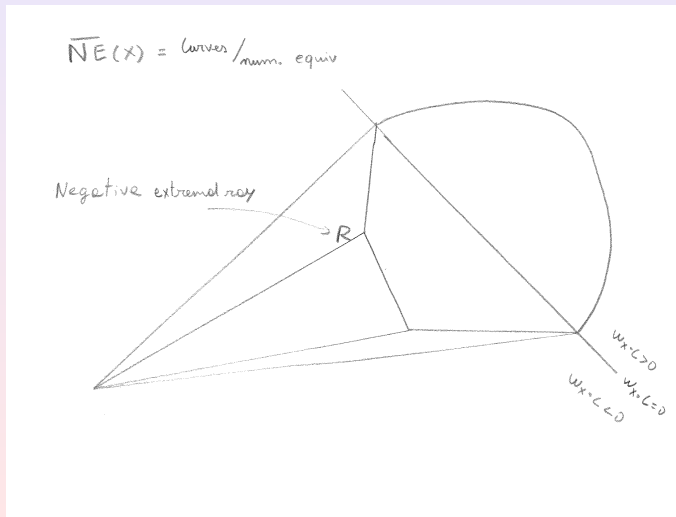
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- By the Cone Theorem, there is a negative extremal ray  $R = \mathbb{R}^+[C]$  where  $C \subset X$  is a curve with  $\omega_X \cdot C < 0$  and there is a morphism  $\text{Cont}_R : X \rightarrow Z$  contracting curves  $[D] \in R$ .

# The Cone Theorem



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- If  $\dim Z = \dim X$  and  $\text{Cont}_R$  is **small** (i.e.  $\dim \text{Exc}(X \rightarrow Z) \leq \dim X - 2$ ) then  $Z$  has “bad” singularities.

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- Instead of replacing  $X$  by  $Z$ , we will replace  $X$  by its flip  $X^+$ .

# Flips

- The **flip** of  $X \rightarrow Z$  is a birational morphism  $X^+ \rightarrow Z$  s. t.
  - 1)  $X \dashrightarrow X^+$  is an isomorphism on the complement of  $\text{Exc}(X \rightarrow Z)$ ,
  - 2)  $\dim \text{Exc}(X^+ \rightarrow Z) \leq \dim(X) - 2$ ,
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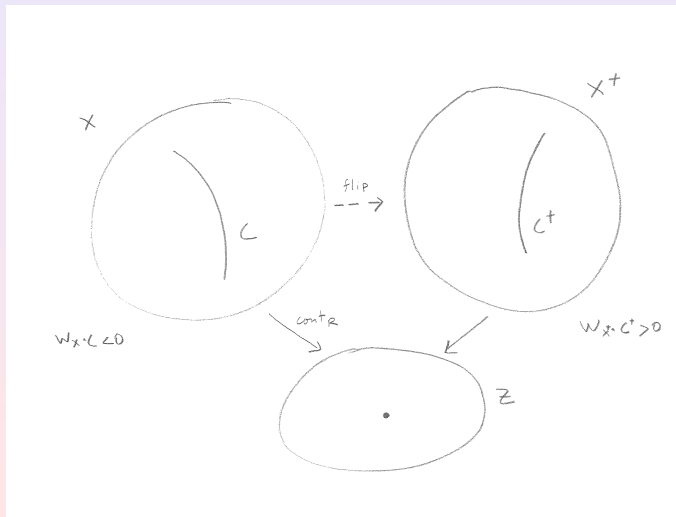
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- In other words, the flip  $X \dashrightarrow X^+$  is a  $\text{codim} \geq 2$  surgery that replaces  $\omega_X$  negative curves by  $\omega_{X^+}$  positive curves.
- If the flip of  $X \rightarrow Z$  exists, it is unique and given by  $\text{Proj}_Z R(\omega_X)$ . So that to construct the flip we must show that  $R(\omega_X)$  is finitely generated over  $Z$  (i.e. if  $Z = \text{Spec} A$ ,  $R(\omega_X)$  is a finitely generated  $A$ -algebra).

# Picture of a flip



# Flips II

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- If  $X \rightarrow Z$  is a divisorial contraction, then  $\rho(Z) = \rho(X) - 1 \geq 0$ . If  $X \dashrightarrow X^+$  is a flip, then  $\rho(X^+) = \rho(X)$ . Therefore, it suffices to show that there is no infinite sequence of flips.

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- In dimension 3, this is relatively easy to do.

# Existence and termination of flips.

## Theorem (Birkar-Cascini-Hacon-M<sup>c</sup>Kernan)

- 1 *Flips exist (in any dimension; for klt pairs).*
- 2 *There are no infinite sequences of flips for the minimal model program with scaling (for any log-big klt pair).*



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Remark: We are unable to show termination of flips. But we can pick a sequence of flips that terminates.

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- (Birkar-Cascini-Hacon-M<sup>c</sup>Kernan) If flips exist in dimension  $n + 1$  (and minimal models exist in dimension  $n$ ) then minimal models exist in dimension  $n + 1$ . (Uses the ideas of the MMP and in particular techniques developed by Shokurov.)

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- By adjunction, we may write  $(K_X + \Delta)|_S = K_S + \Delta_S$  where  $(S, \Delta_S)$  is KLT.
- Shokurov's observation is that to show that  $R(K_X + \Delta)$  is finitely generated over  $Z$  (and hence that the flip exists), it suffices to show that the restricted algebra

$$R_S(K_X + \Delta) := \text{Im}(R(K_X + \Delta) \rightarrow R(K_S + \Delta_S))$$

is finitely generated (over  $Z$ ).



## Existence of flips II

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- Using a Diophantine approximation argument, we show that  $\Theta$  is in fact a  $\mathbb{Q}$ -divisor and  $\Theta = \Theta_m$  for some  $m > 0$ .

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- In this way, we obtain a sequence of rational numbers  $1 \geq t_1 \geq t_2 \cdots \geq 0$  and of flips/div. contractions  $X_i \dashrightarrow X_{i+1}$  such that  $X_i$  is a minimal model for  $(X, \Delta + t_i A)$ .

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- Note that to prove the above result on the finiteness of minimal models, we must use a compactness argument and hence it is critical to work with  $\mathbb{R}$ -divisors.